Unit 5: Transition Metals and Organic Nitrogen Chemistry

IA2 compulsory unit

Externally assessed

Unit description

Introduction

In this unit, the study of electrode potentials builds on the study of redox in Unit 2, including the concept of oxidation number and the use of redox half equations. Students will study further chemistry related to redox, including transition metals.

The organic chemistry section of this unit focuses on arenes and organic nitrogen compounds such as amines, amides, amino acids and proteins. The organic synthesis section requires students to use the knowledge and understanding of organic chemistry that they have gained over the entire specification.

This unit draws on all the other units in the International Advanced Level in Chemistry and students are expected to use their prior knowledge when learning about the areas in this unit. Students will, again encounter ideas of isomerism, bond polarity and bond enthalpy, reagents and reaction conditions, reaction types and mechanisms. Students are also expected to use formulae and balanced equations, and calculate chemical quantities.

Chemistry in action

The study of chemical cells illustrates the impact on scientific thinking when it emerges that ideas developed in different contexts can be shown to be related to a major explanatory principle. In this unit, cell emfs and equilibrium constants are shown to be related to the fundamental criterion for the feasibility of a chemical reaction: the total entropy change.

The explanatory power of the energy-level model for electronic structures is further illustrated by showing how it can help to account for the existence and properties of transition metals. In this context there are opportunities to show the limitations of the models used at this level and to indicate the need for more sophisticated explanations.

Study of the structure of benzene is another example that shows how scientific models develop in response to new evidence. This links to further investigations of the models that chemists use to describe the mechanisms of organic reactions.

The study of catalysts touches on a ‘frontier’ area for current chemical research and development, which is of theoretical and practical importance. This provides an opportunity to show how the scientific community reports and validates new knowledge.

Practical skills	As in previous units, students can begin their practical work in this unit with some simple test-tube reactions, investigating the reactions of transition metal ions in solution. This may lead to an exploration of redox reactions and, therefore, to the core practical on electrochemical cells. Skills in volumetric analysis can be consolidated through titrations for redox systems such as iodine-thiosulfate or manganate(VII) titrations. An opportunity to explore preparative inorganic chemistry is provided in the core practical devoted to making a transition metal complex. In organic chemistry, there are further functional groups to explore and the possibility of preparing an azo dye. The final core practical is an organic synthesis and can be used to showcase a selection of the techniques that students have developed to carry out reactions and purify products efficiently and safely.
Mathematical skills	There are opportunities for the development of mathematical skills in this unit. This includes calculating redox potentials, balancing redox equations from half cells, calculating masses and concentrations from redox titrations, investigating the geometry of transition metal complexes, calculating the resonance stability of benzene from thermodynamic data and calculating percentage yields. (Please see *Appendix 6: Mathematical skills and exemplifications* for more detail.)

Assessment information

- First assessment: June 2020.
- The assessment is 1 hour and 45 minutes.
- The assessment is out of 90 marks.
- Students must answer all questions.
- This paper has three sections:
    - Section A: multiple choice questions
    - Section B: mixture of short-open, open-response, calculations and extended-writing questions
    - Section C: contemporary context question.
- This paper will contain questions that require information from the Data Booklet (see *Appendix 9*).
- This paper will include a minimum of 18 marks that target mathematics at Level 2 or above.
- Students will be expected to apply their knowledge and understanding of experimental methods in familiar and unfamiliar contexts.
- This paper may contain some synoptic questions which require knowledge and understanding from Units 1, 2 and 4.
- Calculators may be used in the examination (see *Appendix 8: Use of calculators*).

Topic 16: Redox Equilibria

Students will be assessed on their ability to:

16.1	understand the terms 'oxidation' and 'reduction' in terms of electron transfer and changes in oxidation number, applied to *s*-, *p*- and *d*-block elements
16.2	know what is meant by the term 'standard electrode potential', $E^{\ominus}$
16.3	know that the standard electrode potential, $E^{\ominus}$, is measured in conditions of: i 298 K temperature ii 100 kPa pressure of gases iii 1.00 mol dm^{-3} concentration of ions
16.4	know the features of the standard hydrogen electrode and understand why a reference electrode is necessary
16.5	understand that different methods are used to measure standard electrode potentials of: i metals or non-metals in contact with their ions in aqueous solution ii ions of the same element with different oxidation numbers
16.6	**CORE PRACTICAL 12** **Investigating some electrochemical cells.**
16.7	be able to calculate a standard emf, $E^{\ominus}_{cell}$, by combining two standard electrode potentials
16.8	be able to write cell diagrams using the conventional representation of half-cells
16.9	understand the importance of the conditions when measuring an electrode potential, *E*
16.10	be able to use standard electrode potentials to predict the thermodynamic feasibility of a reaction
16.11	understand that $E^{\ominus}_{cell}$ is directly proportional to the total entropy change and to ln*K* for a reaction
16.12	understand the limitations of predictions made using standard electrode potentials, in terms of kinetic stability of systems and departure from standard conditions
16.13	know that standard electrode potentials are sometimes referred to as standard reduction potentials and can be listed as an electrochemical series
16.14	understand how standard electrode potentials can be used to predict the thermodynamic feasibility of disproportionation reactions
16.15	be able to carry out both structured and unstructured titration calculations involving redox reactions, including iron(II) ions and potassium manganate(VII) and sodium thiosulfate and iodine
16.16	be able to discuss the uncertainty of measurements and their implications for the validity of the final results

16.17	**CORE PRACTICALS 13a and 13b** **Carry out redox titrations with both:** **i iron(II) ions and potassium manganate(VII)** **ii sodium thiosulfate and iodine**
16.18	understand that fuel cells use the energy released on the reaction of a fuel with oxygen to generate a voltage *Knowledge that methanol and other hydrogen-rich fuels are used in fuel cells is expected.*
16.19	know the electrode reactions that occur in a hydrogen-oxygen fuel cell *Knowledge of hydrogen-oxygen fuel cells with both acidic and alkaline electrolyte is expected.*
	Further suggested practicals: i investigate the percentage of copper in brass, using iodine-thiosulfate titration ii investigate the percentage of iron in iron tablets, using potassium manganate(VII) titration iii prepare crystals of potassium iodate(VII) and measure their purity